

PowerDiffuser: Collaborative Contrastive-Reconstruction Self-Supervised Learning for Robust Power Load Signal Representation

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Abstract—The widespread deployment of smart meters has created significant opportunities for applying artificial intelligence technologies to power system tasks. However, the high cost of data annotation limits the effectiveness of traditional supervised learning in this domain, making self-supervised learning an attractive alternative. In this article, we propose PowerDiffuser, a novel self-supervised learning strategy tailored for power load signals. By leveraging a diffusion model framework, PowerDiffuser integrates two mainstream self-supervised paradigms, namely contrastive learning and reconstruction-based learning, which enables the model to effectively capture both periodic patterns and local features. To address the overfitting issues commonly observed in generic time-series feature extractors when applied to power load tasks, we design two modular spatiotemporal feature extractors specifically engineered to handle samples with varying complexity levels. In addition, we adapt the involution operator to better align with the unique characteristics of power load signals. Extensive experiments on the ISMCBT, ETTh and REDD datasets demonstrate that PowerDiffuser consistently outperforms both time-series general models and existing self-supervised learning strategies across diverse downstream power load tasks. Ablation studies further validate the contributions of the proposed modules and highlight the effectiveness

of transforming 1-D load signals into 2-D periodicity-based representations as a preprocessing step.

Index Terms—Involution, power signal, self-supervised learning, spatiotemporal feature extractors.

I. INTRODUCTION

THE advent of artificial intelligence (AI) technologies has provided novel approaches for power companies to address operational challenges, with extensive research conducted on applications including power load forecasting [1], [2], power generation forecasting [3], subscriber portrait [4], and electricity theft detection [5]. The proliferation of smart meters, intelligent distribution networks, and cloud platforms has enabled power companies to accumulate vast amounts of data, creating favorable conditions for developing effective AI-based power models.

Nevertheless, manual annotation remains indispensable for specific downstream tasks to facilitate supervised learning processes. As demonstrated in [6], electricity theft detection based on power and gas load analysis requires technical experts to explicitly label fraudulent cases. In a similar vein, the study in [4] on household attribute inference from smart meter signals necessitates extensive surveys to obtain accurate labels. These annotation processes often incur substantial costs and face challenges due to privacy concerns and other practical constraints.

Power load signals inherently manifest as sequential data streams characterized by temporal dependencies and dynamic fluctuation patterns. This time-series nature makes them particularly amenable to self-supervised learning (SSL). In the field of AI, SSL significantly reduces the dependency on manually annotated data by exploiting the intrinsic structure and information within the data itself to generate supervisory signals. The two predominant SSL paradigms can be categorized into contrastive learning and reconstruction-based methods. As schematically depicted in Fig. 1, contrastive learning establishes positive and negative sample pairs through carefully designed strategies and learns discriminative feature representations by minimizing the feature space distance between positive sample pairs while maximizing the separation of negative pairs. Fig. 2 illustrates the fundamental framework of reconstruction-based methods, where model is trained to reconstruct complete data from partial or perturbed inputs, thereby capturing the intrinsic structure and essential features of the data. Through optimization of the reconstruction error, the model extracts semantically meaningful

Received 25 February 2025; revised 15 July 2025 and 25 August 2025; accepted 21 October 2025. Date of publication 24 November 2025; date of current version 5 February 2026. This work was supported in part by the Natural Science Foundation of China under Grant 62176193 and Grant 62206204, and in part by the Fundamental Research Funds for the Central Universities, China, under Grant 104972025YJS0118. Paper no. TII-25-1156. (Corresponding author: Cheng Lian.)

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The code is publicly available at <https://github.com/HGYang97/PowerDiffuser>.

Digital Object Identifier 10.1109/TII.2025.3627498

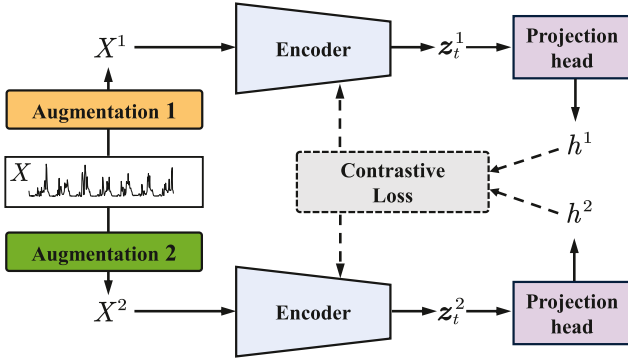


Fig. 1. Generic framework of contrastive learning, where positive and negative sample pairs are defined through augmentation strategies to learn discriminative feature representations by optimizing similarity metrics.

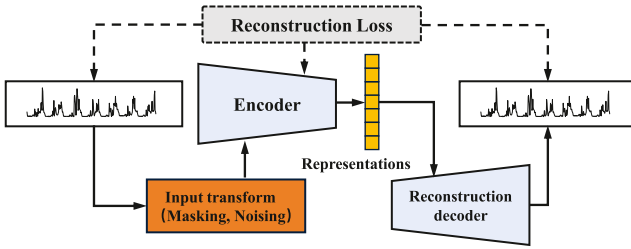


Fig. 2. General paradigm of perturbation-based reconstruction, which trains models to recover clean data from partially masked or noise-corrupted inputs, thereby learning robust feature representations via joint optimization of reconstruction and denoising objectives.

feature representations. Despite these advancements, several critical challenges persist as follows.

- 1) While contrastive learning primarily emphasizes global feature distributions, reconstruction-based methods demonstrate superior capability in extracting local sample characteristics. This inherent limitation poses significant challenges for any single SSL methodology to holistically capture essential data features across diverse downstream tasks in the power domain.
- 2) The effectiveness of both contrastive learning and reconstruction-based approaches exhibits substantial dependence on task formulation. Excessively complex tasks may impede model convergence, whereas overly simplistic formulations often yield inadequate self-supervisory signals.

Due to the significant structural differences between these two methods, few researchers have explored combining them. The emergence of diffusion models has inspired us to design a novel SSL strategy that requires the model to simultaneously perform two tasks: Distinguishing positive and negative sample pairs and interactively generating reconstructions. We believe this design can enhance the model's representational capacity and improve its training robustness. Specifically, when a batch of samples is fed into the model, they undergo two different data augmentation functions, generating two distinct signals. These signals serve as mutual conditions during the diffusion process, cross-reconstructing the augmented representations of each other. After the diffusion process, we also require the classifier to distinguish between positive and negative sample pairs within the batch. We argue that models

trained in this manner can combine the advantages of both paradigms.

In addition, we develop a specialized deep learning architecture tailored to the unique characteristics of power load signals, incorporating two key optimizations. First, building upon previous studies [3], [7] that demonstrate the superior contextual information provided by 2-D representations during reconstruction, and considering recent studies [8], [9] in power applications that show performance improvements through 1-D to 2-D signal transformations, we reformulate the data into a 2-D format, defining it as a multichannel time-series self-supervised task. Second, addressing the overfitting issues observed in general-purpose temporal models [10], [11], [12] when applied to low-complexity power tasks or datasets with limited sampling density, we design two deep learning models based on consistent spatio-temporal interaction principles to accommodate diverse downstream tasks.

Our experimental results demonstrate that the proposed methodology achieves significant performance improvements across various power-related downstream tasks. The specific contributions are as follows.

- 1) We redefine the representation learning process for power load signals as a multichannel time-series self-supervised task and propose an innovative self-supervised learning framework that effectively integrates contrastive learning and reconstruction within the diffusion model paradigm.
- 2) To mitigate overfitting risks, we develop two deep learning architectures based on spatio-temporal interaction principles, specifically designed to handle multiscale task complexity levels and recordings with different but fixed sampling rates, while maintaining compatibility with both self-supervised learning and downstream task fine-tuning.
- 3) Comprehensive experimental evaluations demonstrate that our proposed methodology achieves state-of-the-art performance across evaluation metrics in diverse power-related downstream tasks.

II. RELATED WORK

Power load signals represent a canonical form of time-series data where self-supervised learning effectively enhances representation capability. This paradigm provides an efficient framework for extracting meaningful temporal patterns while reducing label dependency, primarily through two approaches: 1) contrastive learning-based methods leveraging discrimination between positive/negative sample pairs; and 2) reconstruction-based methods targeting future state reconstruction or missing segment imputation. While both demonstrate proven efficacy in temporal representation learning, they differ fundamentally in implementation architectures.

A. Time Series Self-Supervised Methods Based on Contrastive Learning

In the contrastive learning paradigm, CPC [13] constructed positive and negative sample pairs by predicting future time steps, leading to significant advancements in speech classification tasks. InfoTS [14] addressed the challenge of selecting meaningful augmentations for time-series data by leveraging

information theory to adaptively choose optimal augmentations. TCN [15] improved upon this approach by employing a Gaussian distribution-based strategy to constrain the sampling range for context-positive samples. TNC [16] proposed a self-supervised framework for learning generalizable representations of nonstationary time series by leveraging the local smoothness of signals. Using a debiased contrastive objective, TNC ensured that neighboring signals were distinguishable from nonneighboring ones in the encoding space. TS2Vec [17] introduced feature augmentation techniques within hierarchical sample pairs to achieve multigranularity self-supervised representation learning. TS-TCC [18] examined the effects of various feature augmentation methods on contrastive learning performance for time-series signals and proposed a spatiotemporal interaction prediction approach to further enhance model generalization. Although contrastive learning provided effective guidance for model training, performance improvements relied heavily on carefully designed positive and negative sample pairs as well as large batch sizes. This inevitably imposed higher demands on hardware, particularly GPU memory. To address this challenge, we proposed a self-supervised strategy that combined reconstruction methods.

B. Time Series Self-Supervised Methods Based on Reconstruction

The conceptual foundation of reconstruction-based methods originates from Masked Autoencoders (MAE) [19] in computer vision. This approach employs an asymmetric encoder-decoder architecture, where the model's capability to accurately reconstruct randomly masked images serves as an indicator of the encoder's ability to extract deep semantic features. Building upon this foundation, Tarnet [20] introduced a task-guided masking strategy that leverages attention mechanisms to identify and reconstruct critical timestamps, thereby enhancing the perceptual relevance of the reconstruction task. Ti-MAE [21] advanced this paradigm by implementing random masking on embedded time-series data and training an autoencoder for point-level reconstruction, effectively bridging the gap between representation learning and reconstruction methodologies, particularly in forecasting applications. Further innovations were introduced by TimeMAE [22], which shifted the reconstruction target from raw point-level data to latent features while concurrently performing regression tasks. CRT [23] expanded the reconstruction scope by incorporating frequency-domain analysis through fast fourier transform, enabling cross-domain masking and reconstruction of both temporal and spectral signals.

Despite these methodological advancements, reconstruction-based self-supervised strategies inherently present greater challenges compared to contrastive learning approaches. This is particularly evident in time-series analysis, where models struggle to implicitly learn masked region features from 1-D contextual information, as reflected in their reconstruction loss metrics. To address these limitations, we propose a novel approach that directly learns the underlying distribution of time series through a diffusion process, complemented by contrastive interaction constraints, yielding substantial performance improvements.

III. FRAMEWORK

To formally describe the task pipeline, we make the following definitions. Given a dataset $\mathcal{D} = \{X_1, X_2, \dots, X_M\}$, where

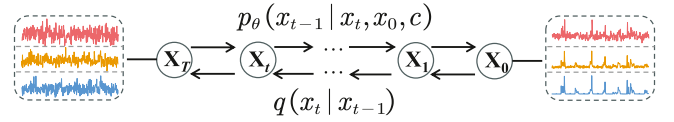


Fig. 3. In diffusion models, the forward process adds noise to time-series data, while the reverse process trains a neural network to remove the noise and recover the original data. During denoising, the model gradually refines the data, using learned knowledge to match real-world patterns.

$X \in \mathbb{R}^L$ represents a power load signal of length L and M denotes the number of samples in the dataset. We reshape the 1-D load signals into 2-D representations during preprocessing. This transformation ensures each channel strictly preserves temporal periodicity, specifically by ensuring that each channel corresponds to weekly or monthly sampling points, thereby maintaining physical meaning and temporal structure consistency. The input data format for our model can thus be expressed as $\hat{X} \in \mathbb{R}^{C \times \frac{L}{C}}$, where C denotes channels.

Our objective is to train model $\mathcal{F}$ using a self-supervised learning strategy to obtain feature representations $F = \mathcal{F}(\hat{X})$ of power load signals. This representation enables effective transfer learning through fine-tuning on downstream tasks, enhancing both generalization capability and robustness.

A. Diffusion Model

As illustrated in Fig. 3, the diffusion model functions through two key processes: a forward noise-adding process and a conditional reverse denoising process. The forward process systematically transforms a clean input $x_0 \sim q(x_0)$ into Gaussian noise $x_T \sim \mathcal{N}(0, I)$ by employing a Markov chain transition. This transition is defined as follows:

$$q(x_t | x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I\right) \quad (1)$$

where $\beta_t \in (0, 1)$ controls the noise schedule at step t . For the reverse process, we extend the standard diffusion to a conditional formulation, where the denoising network predicts the noise component $\epsilon_\phi(x_t, c, t)$, conditioned on task-specific information c

$$p_\theta(x_{t-1} | x_t, c) = \mathcal{N}\left(x_{t-1}; \frac{x_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_\phi(x_t, c, t)}{\sqrt{\alpha_t}}, \beta_t I\right) \quad (2)$$

where $\alpha_t = \prod_{s=1}^t (1 - \beta_s)$. Following DDPM [24], the loss function optimizes a reweighted objective

$$\mathcal{L}(x_0, c) = \sum_{t=1}^T \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} \|\epsilon - \epsilon_\theta(x_t, t, c)\|^2 \quad (3)$$

where T is temporal steps. This conditional mechanism enables targeted refinement of noisy samples toward task-specific distributions.

B. PowerDiffuser

The diffusion model inherently serves as a flexible framework capable of performing reconstruction tasks. By integrating contrastive learning principles, we design a cross-reconstruction process to regularize power signal feature extraction. As illustrated in Fig. 4, PowerDiffuser comprises two branches: the Weak-to-Strong (W2S) and Strong-to-Weak

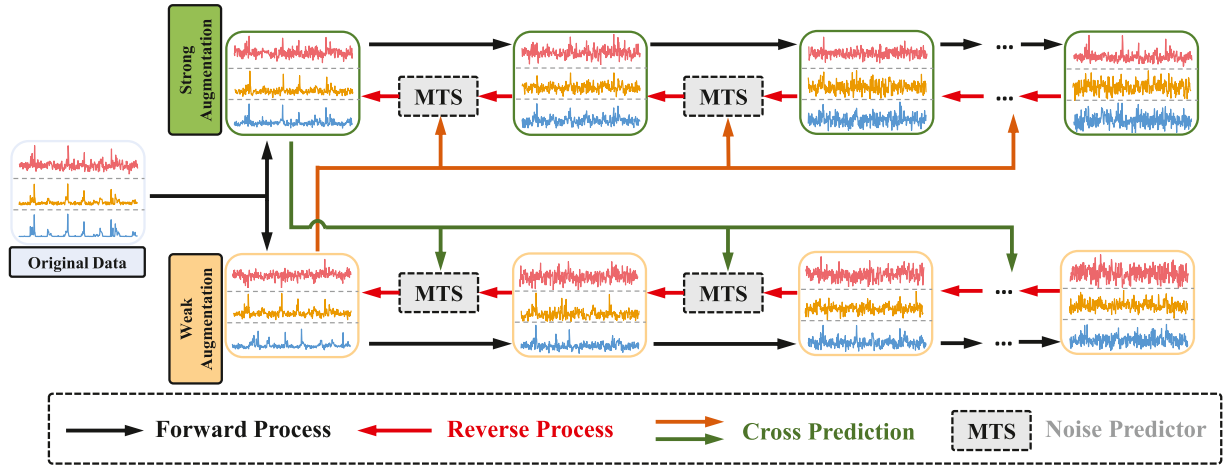


Fig. 4. Framework of the proposed PowerDiffuser, where signals with strong and weak augmentations are cross-reconstructed from each other through the reverse process.

(S2W) augmentation-reconstruction branches. Both W2S and S2W are implemented through conditional DDPMs, where two augmented tensors derived from the original input signal act as mutual conditions for cross-reconstruction. Below, we elaborate on PowerDiffuser from both forward and reverse process perspectives.

Data Augmentation: Contrastive learning, a dominant paradigm in SSL, aligns representations of similar samples (positive pairs) while pushing apart dissimilar ones (negative pairs). In smart meter data analysis, this facilitates distinguishing distinct consumption patterns (e.g., residential vs. industrial) and enhances robustness to noise and temporal shifts. Data augmentation is crucial in constructing meaningful positive and negative pairs: it generates diverse yet semantically consistent variants from the same time-series instance, forming positive pairs, while using sequences from different instances as negatives. Meanwhile, the noise introduced by data augmentation encourages the model to learn invariant representations, enhancing its resilience to perturbations and reducing overfitting, thereby improving reliability in real-world applications.

Forward Diffusion Process: In both branches, the forward diffusion processes operate independently, each resembling standard DDPM. Starting from two augmented tensors $\hat{X}^W$ and $\hat{X}^S$, Gaussian noise is progressively added to generate a sequence of 2-D matrices. Following the findings in [24], [25], we set the diffusion steps T to 1000. This sufficiently large T ensures that both branches converge to identical noise distributions after corruption, a critical prerequisite for initiating cross-reconstruction.

Cross-reconstruct Reverse Process: The reverse process performs a cross-reconstruction task, where both branches start from Gaussian noise and iteratively refine the signals via a shared-parameter neural network for noise prediction and reconstruction. Taking W2S as an example, analogous to conditional DDPM, the neural network incorporates the weakly augmented original signal by concatenating it with the input from the previous diffusion step along the channel dimension. At the t th diffusion step, the noise predictor $\epsilon_\phi(\cdot)$ in the W2S branch takes the reconstructed signal $\hat{X}_t^{S(t)}$ from the previous step, the timestep embedding t , and the condition $\hat{X}^W$ as inputs to estimate the noise component

$$\min \left\| \epsilon - \epsilon_\phi \left(\hat{X}_t^{S(t)}, c = \hat{X}^W, t \right) \right\|_2^2. \quad (4)$$

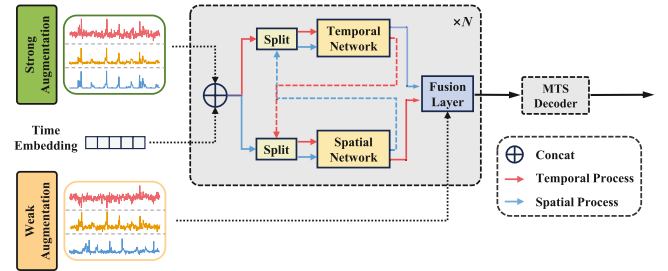


Fig. 5. Detailed computation process of ST-Large. Temporal and spatial network are implemented as one-layer Transformer.

Similarly, the noise predictor in the S2W branch estimates the noise component as

$$\min \left\| \epsilon - \epsilon_\phi \left(\hat{X}_t^{W(t)}, c = \hat{X}^S, t \right) \right\|_2^2. \quad (5)$$

C. Spatiotemporal Neural Network

The purpose of self-supervised training is to obtain a powerful feature extractor that transforms raw power load signals into high-quality representations. During the diffusion process, the denoising network naturally extracts data representations by learning the recovery process at different noise levels. Inspired by previous works [26], we demonstrate that designing feature extractors for two-dimensional power load signals from both spatial and temporal perspectives is crucial. To address the overfitting issue in complex networks, we design two different feature extractors, named ST-Tiny and ST-Large.

ST-Large: As shown in Fig. 5, ST-Large involves two feature extraction processes. Taking W2S as an example, in the encoder part, the temporal network separates input $\in \mathbb{R}^{C \times \frac{L}{C}}$ into a tensor of $\mathbb{R}^{C \times 1 \times K}$ for extracting intraperiod features. Meanwhile, the spatial network separates it into $\mathbb{R}^{1 \times K \times \frac{L}{C}}$ to extract inter-period features at the same timestamp. In the fusion module, we use a basic concat operation to combine the spatial and temporal features and further integrate condition information. During the time embedding process, we adopt the same encoding rules as described in [26]. In the decoder part, we use a Transformer layer to decode the features and generate inputs for the next stage. It is worth noting that in this structure, the number of encoder and

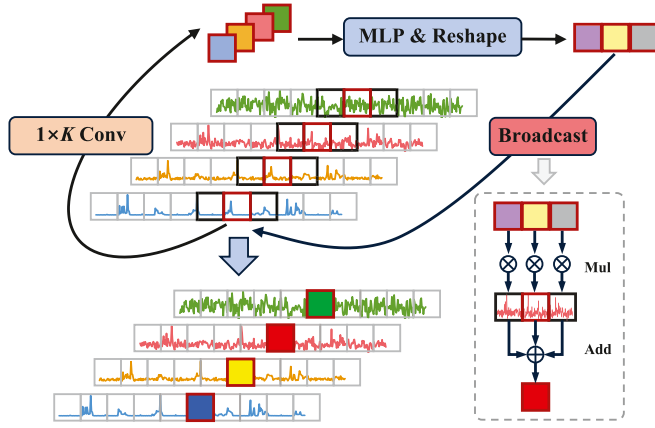


Fig. 6. Detailed computational flow diagram of the 1-D involution operator in ST-Tiny.

Algorithm 1: Feature Transformation.

Require: Input: $\hat{x}_q^i \in \mathbb{R}^{1 \times C}$
Ensure: Output: $\hat{x}_q^{i+1} \in \mathbb{R}^{1 \times C}$

- 1: **procedure** FEATURETRANSFORMATION ($\hat{x}_t^i$)
- 2: Initialize learnable convolution kernel $\theta \in \mathbb{R}^{K \times C}$
- 3: Given window size K , compute windowed features:
- 4: $\hat{x}_{q:K}^i \in \mathbb{R}^{K \times C}$
- 5: Extract intermediate feature vector:
- 6: $G_q^i \in \mathbb{R}^{1 \times C}$
- 7: Project G_q^i via fully-connected layer $W \in \mathbb{R}^{C \times K}$:
- 8: $\hat{G}_q^i = (G_q^i W)^\top \in \mathbb{R}^{K \times 1}$
- 9: Perform broadcasted multiplication and reduction:
- 10: $\hat{x}_q^{i+1} = \sum (\hat{G}_q^i \odot \hat{x}_{q:K}^i)$ ▷Element-wise multiplication followed by summation
- 11: **end procedure**

decoder layers can be flexibly stacked to meet the requirements of extracting power load signals of varying complexity.

ST-Tiny: The involution operator [27] is a computation process in the computer vision domain that can replace convolution, commonly considered to have spatial specific and channel agnostic. We modified its computation process to generalize it to temporal data and believe that it can simultaneously handle spatiotemporal features in temporal data, as shown in Fig. 6. Similar to ResNet-1D [28], the computation traverses multi-channel temporal signals along the time dimension. Taking W2S as an example, the original input to the computation at the i th involution layer for timestamp q is represented by $\hat{x}_q^i \in \mathbb{R}^{1 \times C}$. The specific forward computation is shown in Algorithm 1.

D. Loss Functions

The loss construction in the pre-training process is divided into three components: diffusion process, reconstruction, and contrastive learning. The final loss function is the sum of these three parts.

Diffusion Process: We train the feature extractors of the W2S and S2W branches using the noise prediction loss defined in (4) and (5). The diffusion objective is equivalent to denoising score matching; it shapes representations under multilevel noise, captures the local geometry of the data manifold, and improves robustness.

Reconstruction: We use the MSE loss to compare the reconstructed outputs with their corresponding augmented targets. Specifically, $\hat{X}_T^{S(r)}$ denotes the final denoised output in the W2S branch, which is reconstructed from Gaussian noise conditioned on the weakly augmented signal $\hat{X}^W$. Similarly, $\hat{X}_T^{W(r)}$ is reconstructed in the S2W branch conditioned on $\hat{X}^S$. The reconstruction loss is defined as

$$\mathcal{L}_R = \frac{1}{|\Omega|} \sum_{u \in \Omega} \left(\hat{X}_{T,u}^{S(r)} - \hat{X}_u^S \right)^2 + \frac{1}{|\Omega|} \sum_{u \in \Omega} \left(\hat{X}_{T,u}^{W(r)} - \hat{X}_u^W \right)^2 \quad (6)$$

where Ω denotes the set of element indices of the signal tensors. The reconstruction loss constrains the model to retain sufficient information to faithfully reconstruct the input, preventing purely discriminative training from discarding excessive information.

Contrastive Learning: Given a batch of H input samples, we apply strong and weak augmentations and perform denoising at $T/2$ and T steps, resulting in a total of $6H$ potential samples. Among them, the 6 samples derived from the same original sample are considered positive pairs, while samples from different original inputs are treated as negative pairs. We employ the Normalized Temperature-scaled Cross-Entropy Loss to optimize the model, ensuring that the representations of the same sample variations are closer while keeping them distinct from other samples. This design extends the loss formulation and the construction of positive and negative sample pairs as proposed in TS-TCC [18] (<https://github.com/emadeldeen24/TS-TCC>). The loss function is formulated as

$$\mathcal{L}_C = - \sum_{i=1}^H \sum_{a=1}^6 \sum_{b=1, b \neq a}^6 \log \frac{\exp(\text{sim}(h_{i,a}, h_{i,b})/\tau)}{\sum_{j=1}^H \sum_{k=1}^6 \exp(\text{sim}(h_{i,a}, h_{j,k})/\tau)} \quad (7)$$

where

- 1) $h_{i,a}$ and $h_{i,b}$ are feature representations obtained from a neural network encoder and are ℓ_2 -normalized.
- 2) $\text{sim}(h_{i,a}, h_{i,b}) = h_{i,a}^\top h_{i,b}$ represents the cosine similarity.
- 3) τ is a temperature hyperparameter that controls the scale of contrastive loss, and it is typically set to 0.1 in our experiments.

The contrastive loss provides a mutual-information lower-bound objective, encouraging embeddings of positive signals to align while pushing apart negative signals.

Overall, within the diffusion-model framework, the three loss terms cooperate to form a complementary objective that enhances discriminability, preserves fidelity, and captures the data manifold.

E. Theoretical Principles

We discussed the advantages of the PowerDiffuser framework from the perspectives of mutual information maximization and information bottleneck theory, and explained why the combination of reconstruction and contrastive learning is effective.

1) Mutual Information Maximization: The essence of self-supervised learning is to maximize the mutual information between the input X and its latent representation Z , i.e., $I(X; Z)$, so that Z contains sufficient information to reconstruct X or predict its attributes.

- a) **Reconstruction Learning** directly optimizes $I(X; Z)$ by minimizing the reconstruction loss $\mathcal{L}_R$, which enhances the encoding ability of Z with respect to X .
- b) **Contrastive Learning** implicitly improves $I(Z; Y)$ by encouraging the alignment of positive sample pairs, where Y denotes the target variable of the downstream task. Specifically:
 - i) The contrastive objective enforces the similarity between the clean and augmented views (e.g., different noisy versions) of the same input, Z_{clean} and Z_{aug} , which approximates the maximization of the mutual information $I(Z_{\text{clean}}; Z_{\text{aug}})$.
 - ii) Based on the information inequality $I(Z; Y) \geq I(Z_{\text{clean}}; Z_{\text{aug}})$, contrastive loss can be seen as a lower bound of task-relevant mutual information.

From an information-theoretic perspective, the total loss of PowerDiffuser can be approximated as

$$\mathcal{L}_{\text{total}} \approx -I(X; Z) - \lambda \cdot I(Z_{\text{clean}}; Z_{\text{aug}}). \quad (8)$$

By jointly optimizing these two objectives, PowerDiffuser effectively balances signal reconstruction and representation invariance, leading to improved generalization in downstream tasks. The combination of both terms provides a decoupled yet complementary view of the information flow: reconstruction promotes informativeness, while contrastive learning enhances consistency under local perturbations or noise.

Due to the long-term dependencies in power load signals, representations are required to capture cross-temporal dependencies. Contrastive learning achieves this by enforcing temporal invariance constraints, while reconstruction ensures the temporal continuity of the time series.

2) Information Bottleneck Theory: The information bottleneck theory aims to learn a compact representation Z from the input X , such that the retained information is maximally relevant to the target task Y , while minimizing the irrelevant or nuisance information. This leads to an optimization tradeoff between $I(Z; Y)$ and $I(Z; X)$.

- a) $\mathcal{L}_R$ enforces the preservation of complete information about X in Z (high $I(X; Z)$), but it may include a large amount of task-irrelevant noise or redundant features (e.g., background, disturbances).
- b) $\mathcal{L}_C$ enhances task-relevant features by encouraging invariance under augmentation (e.g., noise injection, temporal cropping), thus increasing $I(Z; Y)$ and filtering out nuisance variables unrelated to the downstream task.

Joint optimization of reconstruction loss $\mathcal{L}_R$ and contrastive loss $\mathcal{L}_C$ aligns with the IB principle

$$\min_Z [I(Z; X) - \beta I(Z; Y)]$$

where $\mathcal{L}_R$ constrains the lower bound of $I(Z; X)$ to avoid information loss of critical features; $\mathcal{L}_C$ increases $I(Z; Y)$ to enhance discriminability; β balances the trade-off and can be interpreted as a compression coefficient larger β , stronger compression.

The nonstationarity of power load signals, such as periodic fluctuations and sudden noise, requires that the learned representations preserve both global pattern completeness (ensured by reconstruction-based objectives) and local perturbation invariance (guaranteed by contrastive objectives). For instance, augmentation strategies in contrastive learning simulate transient disturbances that occur in real-world power grids, while

TABLE I
ISMCBT DATASET PARTITIONING AND BRIEF INTRODUCTION

No.Que	Question	Answers	Users
300	Age of chief income earner	Young (<35)	436
		Medium (35-65)	2819
		Old (>65)	953
310	Chief income earner has retired or not	Yes	1285
		No	2947
450	House type	Apartment or bungalow	1123
		Detached	1690
		Semi-detached or terraced	1310
4704	Cooking facility type	Electrical	1272
		Not Electrical	2960

reconstruction learning ensures that the underlying periodic trends are maintained.

IV. EXPERIMENTAL SETUP

A. Datasets

The Irish Smart Meter Customer Behavior Trial (ISMCBT) dataset, curated by Ireland's energy regulatory authority, comprises smart meter recordings from 4,232 residential households over 536 days at 30-min resolution, alongside responses to over 200 survey questions per subscriber. We select classification tasks numbered 300, 310, 450, and 4704 as detailed in Table I.

The ETTh dataset contains hourly-sampled power load records from Chinese substations, serving as an established benchmark for multivariate time-series forecasting. We extend its application to evaluate imputation and short-term forecasting task performance in this study.

The REDD dataset comprises electricity consumption records from six U.S. households, capturing both whole-home and appliance-level power usage across over ten types of devices. Initially designed for energy disaggregation research, it is repurposed in this study to evaluate anomaly detection performance under realistic residential load conditions.

B. Data Processing

We conducted temporal segmentation on all datasets to simulate power load signals with varying observation durations. For the ISMCBT dataset, we established two temporal scales: weekly segmentation (ISMCBT-Tiny, 336 sampling points restructured into 7×48 matrices) and monthly segmentation (ISMCBT-Large, 1440 points structured as 30×48 matrices), where the monthly partitioning adopted 30-day cycles for standardization. The ETTh dataset underwent dedicated preprocessing for imputation and short-term time-series forecasting task training, implementing two granularities: ETTh-Tiny with 7×24 configuration (168 points) and ETTh-Large with 30×24 structure (720 points). All datasets were normalized and sample interpolation was used to account for sparse missing values. The matrix transformation explicitly preserves temporal topology, where row dimensions represent consecutive days and column dimensions correspond to fixed measurement intervals (48 daily readings for ISMCBT, 24 for ETTh), maintaining the intrinsic cyclicity of power consumption patterns.

TABLE II
COMPARISON OF CLASSIFICATION TASKS ON THE ISMCBT-TINY DATASET

Method	Task300			Task310			Task450			Task4704		
	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC
SOTA Time Series General Models												
Autoformer	0.673	0.588	0.687	0.716	0.634	0.724	0.405	0.392	0.683	0.705	0.682	0.778
ModernTCN	0.724	0.645	0.736	0.776	<u>0.712</u>	0.796	0.437	0.432	0.722	0.760	0.745	0.812
Informer	0.738	0.657	0.751	<u>0.777</u>	0.711	0.798	0.443	0.440	0.728	0.767	0.750	0.813
ATFNet	0.684	0.599	0.702	0.741	0.673	0.768	0.421	0.414	0.709	0.721	0.698	0.789
Self-Supervised Learning (Linear Probing)												
CPC	0.691	0.611	0.705	0.679	0.611	0.745	0.469	0.468	0.754	0.750	0.734	0.807
TS2Vec	0.695	0.614	0.709	0.707	0.638	0.751	0.479	0.477	0.763	0.752	0.737	0.809
TS-TCC	0.692	0.612	0.707	0.718	0.648	0.760	0.487	0.486	0.771	0.747	0.731	0.805
MAE-1D	0.642	0.559	0.655	0.657	0.588	0.708	0.403	0.398	0.682	0.649	0.632	0.723
TimeMAE	0.730	0.651	0.742	0.742	0.672	0.786	0.521	0.518	0.802	0.772	0.758	0.834
CRT	0.737	0.658	0.749	0.744	0.676	0.790	0.521	0.519	0.803	0.796	0.782	0.857
PowerDiffuser-Tiny	0.743	0.664	0.754	0.770	0.702	0.812	0.534	0.534	0.821	0.796	0.783	0.856
PowerDiffuser-Large	0.712	0.631	0.723	0.714	0.642	0.762	0.492	0.491	0.795	0.668	0.656	0.762
Self-Supervised Learning (Fine-Tuning)												
CPC	0.719	0.639	0.730	0.742	0.672	0.786	0.480	0.478	0.771	0.771	0.757	0.833
TS2Vec	0.721	0.641	0.732	0.744	0.675	0.789	0.485	0.484	0.777	0.778	0.764	0.841
TS-TCC	0.722	0.642	0.734	0.738	0.668	0.784	0.509	0.508	0.797	0.779	0.766	0.842
MAE-1D	0.653	0.571	0.667	0.716	0.646	0.769	0.421	0.416	0.710	0.661	0.647	0.752
TimeMAE	<u>0.759</u>	<u>0.679</u>	0.770	0.764	0.698	0.810	0.566	0.565	0.851	<u>0.812</u>	<u>0.798</u>	<u>0.870</u>
CRT	0.758	<u>0.679</u>	<u>0.772</u>	0.776	0.710	<u>0.822</u>	<u>0.564</u>	<u>0.564</u>	0.849	0.774	0.760	0.842
PowerDiffuser-Tiny	0.781	0.705	0.791	0.789	0.724	0.835	<u>0.564</u>	0.565	<u>0.850</u>	0.819	0.808	0.885
PowerDiffuser-Large	0.715	0.635	0.728	0.715	0.645	0.765	0.491	0.491	0.794	0.693	0.682	0.789

Bold indicates the optimal choice, while underline denotes the suboptimal alternative.

Since the REDD dataset lacks ground-truth anomaly labels, we follow common practice by injecting synthetic anomalies to facilitate the evaluation of unsupervised anomaly detection methods. Specifically, we introduce four types of anomalies into the aggregated power consumption data: 1) spike anomalies, where sudden surges are simulated by amplifying values by a factor of 2.5–4.0; 2) dropout anomalies, created by scaling values down to 1%–10% of their original magnitude; 3) noise bursts, in which localized Gaussian noise is added over short windows; 4) flat segments, where values are held constant to mimic sensor faults or frozen readings. The above noise injection method adopts equal-weight random sampling. Anomalies are randomly injected into 5% and 10% of the total time steps to ensure sparsity and realism.

C. Implementation Details

In the experiments, pretraining was conducted for 120 epochs, and downstream tasks were trained for 80 epochs. A batch size of 64 was used. The weights of all learnable parameters in the models were initialized using Kaiming initialization and optimized with the Adam optimizer. The initial learning rate was set to 3×10^{-4} , with a weight decay of 1×10^{-5} . All experiments were implemented using PyTorch and trained on an NVIDIA GeForce RTX 4090 Ti GPU.

In terms of model implementation, considering the time cost, ST-Large adopts a structure with one encoder layer and one decoder layer. ST-Tiny utilizes four involution blocks with $K = 3$ as the feature extraction units. For downstream tasks, we selected the model at diffusion step $T = 5$ for both linear probing and fine-tuning, and demonstrated its superiority in ablation experiments. In the contrastive learning loss, we designed a

two-layer MLP to extract feature representations for $h_{i,a}$ and $h_{i,b}$.

V. EXPERIMENT RESULTS

A. Downstream Task Experiments

1) *Classification Task*: Table II and Table III present the performance of PowerDiffuser on classification tasks, comparing it with both time-series general models and self-supervised strategies under linear probing and fine-tuning settings. PowerDiffuser demonstrates superior classification performance across most tasks, owing to its integration of contrastive learning and reconstruction, which effectively captures intricate local signal patterns and periodic distributions, necessitating robust feature extraction capabilities.

Under fine-tuning, the Tiny version remains competitive across all tasks on the ISMCBT-Tiny dataset, outperforming the second-best model by an average AUC margin of 0.016 across task300, task310, and task4704. The Large version dominates ISMCBT-Large, surpassing the second-best model by an average AUC margin of 0.004 across task300, task450, and task4704. Under linear probing, the Tiny version also performs competitively on ISMCBT-Tiny, achieving leading or comparable results in most tasks, with particularly strong performance on task4704. On ISMCBT-Large, the Large version consistently ranks first or second across all four tasks. These results demonstrate the model's strong transferability and representational capacity, even under limited supervision.

In addition, two key observations emerge: 1) the experiment highlights the universality of overfitting in the representation learning process of power load signals. For instance, general time-series models and ST-Large models, which have larger

TABLE III
COMPARISON OF CLASSIFICATION TASKS ON THE ISMCBT-LARGE DATASET

Method	Task300			Task310			Task450			Task4704		
	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC
SOTA Time Series General Models												
Autoformer	0.704	0.619	0.721	0.769	0.689	0.782	0.447	0.441	0.745	0.738	0.712	0.807
ModernTCN	0.731	0.643	0.743	0.780	0.726	0.804	0.452	0.447	0.756	0.773	0.752	0.822
Informer	0.745	0.657	0.758	0.783	0.731	0.807	0.451	0.446	0.755	0.771	0.749	0.819
ATFNet	0.715	0.625	0.729	0.773	0.710	0.798	0.449	0.444	0.753	0.741	0.718	0.814
Self-Supervised Learning (Linear Probing)												
CPC	0.703	0.614	0.715	0.665	0.621	0.742	0.482	0.478	0.770	0.762	0.738	0.828
TS2Vec	0.710	0.621	0.719	0.722	0.659	0.770	0.493	0.488	0.784	0.763	0.743	0.830
TS-TCC	0.711	0.622	0.721	0.733	0.668	0.783	0.502	0.497	0.799	0.759	0.739	0.826
MAE-1D	0.656	0.563	0.666	0.671	0.615	0.726	0.418	0.413	0.712	0.662	0.640	0.730
TimeMAE	0.741	0.655	0.748	0.759	0.703	0.803	0.535	0.531	0.818	0.786	0.764	0.844
CRT	0.750	0.664	0.756	0.762	0.707	0.809	0.536	0.531	0.820	0.812	0.792	0.871
PowerDiffuser-Tiny	0.785	0.701	0.791	0.769	0.713	0.812	0.543	0.539	0.827	0.799	0.781	0.857
PowerDiffuser-Large	<u>0.792</u>	<u>0.709</u>	<u>0.798</u>	0.774	0.719	0.816	0.564	0.561	<u>0.848</u>	0.801	0.785	0.861
Self-Supervised Learning (Fine-Tuning)												
CPC	0.719	0.630	0.726	0.743	0.687	0.787	0.480	0.475	0.768	0.769	0.749	0.831
TS2Vec	0.721	0.632	0.728	0.744	0.690	0.790	0.485	0.480	0.774	0.778	0.760	0.839
TS-TCC	0.722	0.634	0.729	0.739	0.685	0.786	0.509	0.505	0.794	0.779	0.762	0.840
MAE-1D	0.661	0.571	0.671	0.642	0.592	0.700	0.417	0.412	0.710	0.659	0.640	0.750
TimeMAE	0.758	0.671	0.765	0.764	0.711	0.810	<u>0.566</u>	<u>0.562</u>	0.847	0.811	0.791	0.866
CRT	0.763	0.678	0.769	0.806	0.749	0.851	0.563	0.559	0.845	0.807	0.789	0.866
PowerDiffuser-Tiny	0.782	0.697	0.788	0.784	0.731	0.829	0.564	0.560	0.847	<u>0.819</u>	<u>0.802</u>	<u>0.882</u>
PowerDiffuser-Large	0.795	0.712	0.801	<u>0.799</u>	<u>0.742</u>	<u>0.844</u>	0.572	0.568	0.856	0.820	0.803	0.883

Bold indicates the optimal choice, while underline denotes the suboptimal alternative.

TABLE IV
COMPARISON OF IMPUTATION TASKS ON THE ISMCBT AND ETTh DATASET

Method	ISMCBT-Large			ISMCBT-Tiny			ETTh-Large			ETTh-Tiny		
	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE
SOTA Time Series General Models												
Autoformer [12]	0.0840	0.1815	9.82%	0.1397	0.2355	12.43%	0.0463	0.1360	7.34%	0.1168	0.2140	10.79%
ModernTCN [29]	0.0616	<u>0.1542</u>	<u>8.35%</u>	0.0831	0.1712	8.98%	0.0427	0.1298	6.99%	0.0872	0.1851	9.45%
Informer [11]	0.0815	0.1762	9.45%	0.1157	0.2280	11.97%	0.0375	<u>0.1242</u>	<u>6.71%</u>	0.1199	0.2176	10.96%
ATFNet [10]	0.0763	0.1675	8.99%	0.1392	0.2402	12.65%	0.0383	0.1267	6.83%	0.1402	0.2510	12.64%
Self-Supervised Learning (Linear Probing)												
CPC [13]	0.1310	0.2397	12.96%	0.1382	0.2453	12.95%	0.1054	0.2178	11.79%	0.0925	0.2031	10.89%
TS2Vec [17]	0.1240	0.2308	12.49%	0.1309	0.2364	12.50%	0.0890	0.2102	11.38%	0.0786	0.1902	10.25%
TS-TCC [18]	0.1182	0.2260	12.24%	0.1183	0.2257	11.94%	0.0808	0.2054	11.13%	0.0816	0.1963	10.58%
MAE-1D [19]	0.2061	0.3352	18.09%	0.1919	0.3205	17.22%	0.1900	0.3300	18.00%	0.1897	0.3275	17.66%
TimeMAE [22]	0.0997	0.1995	10.78%	0.0949	0.1933	10.40%	0.0652	0.1821	9.87%	0.0634	0.1772	9.62%
CRT [23]	0.0922	0.1923	10.42%	0.0917	0.1905	10.28%	0.0596	0.1752	9.48%	0.0613	0.1731	9.40%
PowerDiffuser-Tiny	0.0821	0.1817	9.82%	0.0897	0.1872	10.10%	0.0531	0.1680	9.11%	0.0559	0.1693	9.18%
PowerDiffuser-Large	0.0780	0.1769	9.57%	0.1126	0.2116	11.39%	0.0516	0.1628	8.83%	0.1014	0.2060	11.15%
Self-Supervised Learning (Fine-Tuning)												
CPC [13]	0.1006	0.2010	10.87%	0.1075	0.2083	11.22%	0.0649	0.1867	10.11%	0.0723	0.1907	10.32%
TS2Vec [17]	0.0937	0.1940	10.50%	0.1002	0.2005	10.80%	0.0584	0.1793	9.72%	0.0539	0.1690	9.18%
TS-TCC [18]	0.0875	0.1872	10.13%	0.0878	0.1894	10.20%	0.0506	0.1712	9.27%	0.0508	0.1721	9.32%
MAE-1D [19]	0.1558	0.2815	15.20%	0.1404	0.2678	14.43%	0.1388	0.2790	15.15%	0.1383	0.2770	15.02%
TimeMAE [22]	0.0683	0.1660	8.98%	0.0634	0.1610	8.71%	<u>0.0339</u>	0.1305	7.06%	0.0318	0.1261	6.82%
CRT [23]	0.0605	0.1574	8.51%	<u>0.0601</u>	<u>0.1553</u>	<u>8.40%</u>	0.0412	0.1449	7.84%	0.0375	0.1402	7.60%
PowerDiffuser-Tiny	<u>0.0604</u>	0.1567	8.48%	0.0582	0.1530	8.26%	0.0416	0.1452	7.86%	<u>0.0327</u>	<u>0.1308</u>	<u>7.09%</u>
PowerDiffuser-Large	0.0563	0.1505	8.14%	0.0713	0.1744	9.43%	0.0302	0.1230	6.67%	0.0419	0.1455	7.88%

Bold indicates the optimal choice, while underline denotes the suboptimal alternative.

TABLE V
COMPARISON OF SHORT-TERM TIME-SERIES FORECASTING TASK ON THE ETTH DATASET

Method	Horizon=48		Horizon=96	
	MSE	MAE	MSE	MAE
SOTA Time Series General Models				
Autoformer [12]	0.175	0.352	0.192	0.369
ModernTCN [29]	0.159	0.335	<u>0.189</u>	0.362
Informer [11]	<u>0.164</u>	<u>0.330</u>	0.193	0.363
ATFNet [10]	0.168	0.340	0.197	0.374
Self-Supervised Learning (Linear Probing)				
CPC [13]	0.166	0.336	0.197	0.367
TS2Vec [17]	0.158	0.339	0.194	0.368
TS-TCC [18]	<u>0.156</u>	0.335	0.191	0.363
MAE-1D [19]	0.231	0.433	0.242	0.448
TimeMAE [22]	<u>0.156</u>	0.323	0.192	0.362
CRT [23]	0.159	0.324	0.191	0.361
PowerDiffuser-Tiny	0.158	0.326	0.190	0.362
PowerDiffuser-Large	0.166	0.333	0.197	0.373
Self-Supervised Learning (Fine-Tuning)				
CPC [13]	0.159	0.326	0.194	0.362
TS2Vec [17]	0.157	0.331	<u>0.189</u>	<u>0.360</u>
TS-TCC [18]	<u>0.156</u>	0.328	0.186	0.358
MAE-1D [19]	0.169	0.340	0.202	0.389
TimeMAE [22]	0.155	0.331	<u>0.189</u>	0.361
CRT [23]	<u>0.156</u>	<u>0.322</u>	<u>0.189</u>	0.362
PowerDiffuser-Tiny	0.155	0.318	0.186	0.361
PowerDiffuser-Large	0.162	0.331	0.186	0.363

Bold indicates the optimal choice, while underline denotes the suboptimal alternative.

TABLE VI
COMPARISON OF ANOMALY DETECTION TASK ON THE REDD DATASET

Method	Injection Rate=5%		Injection Rate=10%	
	ACC	F1	ACC	F1
Autoformer [12]	0.953	0.812	0.962	0.873
ModernTCN [29]	0.961	0.843	0.969	0.896
Informer [11]	0.952	0.801	0.960	0.872
ATFNet [10]	0.964	0.856	0.964	0.893
PowerDiffuser-Tiny	<u>0.971</u>	0.881	0.961	0.891
PowerDiffuser-Large	0.977	<u>0.879</u>	<u>0.967</u>	<u>0.893</u>

Bold indicates the optimal choice, while underline denotes the suboptimal alternative.

parameter sizes, exhibit notable overfitting on the ISMCBT-Tiny dataset but perform better on ISMCBT-Large. This pattern is similarly observed in time-series self-supervised models. Since these models typically employ simpler and modular architectures, their performance shows smaller variations between tiny and large datasets, underscoring the necessity of designing power load models tailored to the specific characteristics of the sampled data; 2) we observed that in self-supervised strategies based on power load signals, reconstruction-based approaches generally outperform those based on contrastive learning. This suggests that power load signals prioritize extracting information from the same time step across local or different cycles, particularly in scenarios with missing data or disturbances.

2) Imputation Task: Table IV presents the results of the imputation task. To demonstrate the generalization capability of

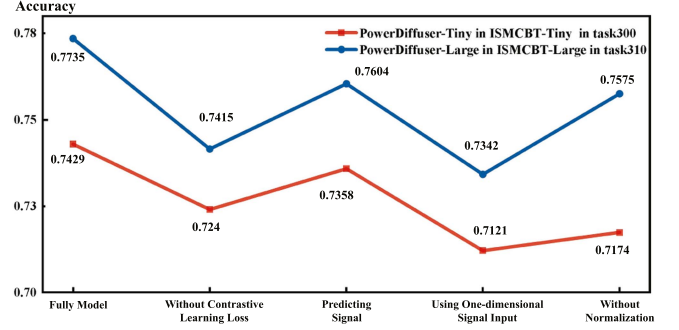


Fig. 7. Impact of modifying or removing certain settings on the performance of the PowerDiffuser model. The numbers in the figure represent the accuracy of the classification task.

the method, we conducted experiments on both the ISMCBT and ETTh datasets. The overall findings align with those of the classification task. Models with large parameters still exhibit overfitting on the tiny dataset but perform well on the large dataset. Our proposed model consistently achieves optimal or suboptimal results across all settings, with PowerDiffuser-Large achieving the lowest MSE on ISMCBT-Large and ETTh-Large, while PowerDiffuser-Tiny performs best on ISMCBT-Tiny. Although performance on ETTh-Tiny is competitive, TimeMAE achieves the best overall result in this setting.

3) Short-Term Time Series Forecasting Task: Table V compares the forecasting performance of all models on the ETTh-Tiny dataset under two short-term horizons (48 and 96). The results demonstrate that PowerDiffuser-Tiny exhibits superior performance in this task. Compared with other self-supervised baselines, TS-TCC and TimeMAE also perform well, especially in the fine-tuning setting. Overall, short-term forecasting is a relatively simple downstream task, and as model size increases, current methods tend to achieve comparable performance.

4) Anomaly Detection Task: We evaluated the performance of various models under different anomaly injection rates. As shown in Table VI, PowerDiffuser demonstrates significant advantages in the anomaly detection task. Notably, under a low injection rate of 5%, the PowerDiffuser-Tiny model achieves the best performance with an accuracy of 0.971 and an F1 score of 0.881. When the anomaly injection rate increases to 10%, the ModernTCN model excels, reaching an accuracy of 0.969 and an F1 score of 0.896. This suggests that its short-range feature extraction structure is particularly effective for capturing high-frequency anomalies. The results also indicate that although traditional time-series models yield relatively high accuracy, their F1 scores are comparatively lower, reflecting limited capability in detecting minority-class samples. Since anomaly detection is an unsupervised process, its performance heavily relies on the modeling capacity of the feature extractor. Therefore, we argue that ST is more suitable for capturing patterns in power load signals.

B. Ablation Experiments

1) Deconstructing PowerDiffuser: To demonstrate the contribution of different components in PowerDiffuser to accuracy improvement, we systematically deconstructed its design and evaluated it on two classification tasks, as shown in Fig. 7. To better highlight the capabilities of the self-supervised process, we adopted the linear probing approach for downstream tasks.

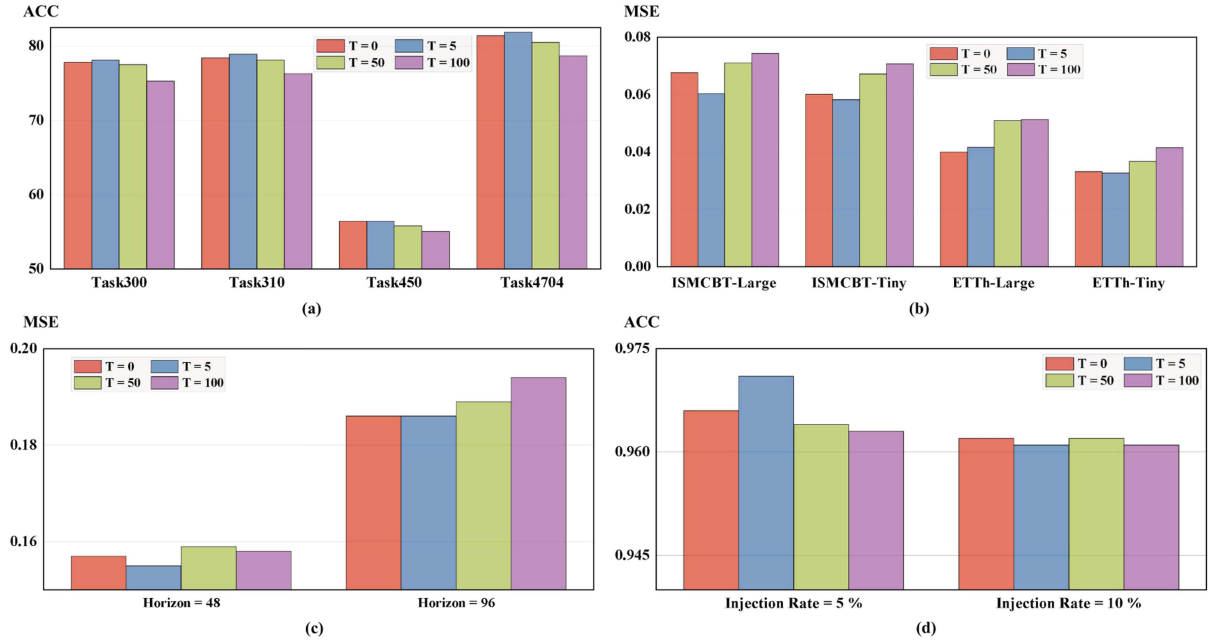


Fig. 8. Impact of models pretrained with different diffusion steps on downstream task performance. (a) Classification task. (b) Imputation task. (c) Short-term time-series forecasting task. (d) Anomaly detection task.

TABLE VII
PERFORMANCE COMPARISON BETWEEN POWERDIFFUSER AND
NONDIFFUSION HYBRID MODEL

Method	Accuracy	F1-score	AUC
MAE + SimCLR (no diffusion)	0.7129	0.6577	0.7618
PowerDiffuser (ours)	0.7896	0.7244	0.8350

Bold indicates the optimal choice.

- Contrastive Learning Loss Removal*: Removing the contrastive learning loss and relying solely on reconstruction demonstrated the importance of contrastive constraints.
- Prediction Target Modification*: Changing the diffusion process target from predicting noise to predicting the original signal surprisingly resulted in worse performance, aligning with findings in [25].
- 2-D Signal Conversion*: Converting power load signals into 2-D representations proved crucial, as it provided both spatial and temporal features, enabling the model to better capture underlying sample information.
- Normalization Removal*: Eliminating normalization significantly degraded performance, confirming its necessity as a preprocessing step for power load signals.

2) Ablation on Pretrained Models With Different Diffusion Steps T : We evaluate the impact of pretraining with different diffusion steps on downstream performance across four tasks, using PowerDiffuser-Tiny in all experiments. As shown in Fig. 8, the model achieves the best performance on most tasks when the diffusion step is set to $T = 5$. We believe that allowing a moderate level of noise during pretraining enables more flexible fine-tuning in downstream tasks, which is consistent with the findings in [25].

3) Ablation on Different Feature Augmentation Combinations: Designing data augmentation methods tailored to specific

data characteristics is a critical factor in the success of contrastive learning. We systematically explored various feature augmentation combinations to identify the optimal strategy for power load signals, with the results illustrated in Fig. 9. Under the linear probing setting, we observed that overly aggressive augmentations, such as random noise, rendered the reconstruction process excessively challenging, as demonstrated in Fig. 9(a).

From a loss perspective, combinations incorporating random mask methods achieved the lowest loss values. From a downstream task performance perspective (ISMCBT-Tiny, Task310), the combination of permutation and random mask emerged as the most effective. Consequently, we adopted this combination as the default feature augmentation strategy for PowerDiffuser. This exploration underscores the importance of aligning the difficulty of self-supervised strategy design with both model size and sample complexity.

4) Effect of Diffusion Framework: To isolate the contribution of the diffusion mechanism in our contrastive-reconstruction fusion framework, we construct an alternative hybrid baseline that does not rely on a diffusion model. Specifically, we implement a dual-branch encoder structure inspired by MAE [19] and SimCLR [30], where one branch performs masked autoencoding for reconstruction, and the other branch applies contrastive learning with augmentations. Both branches share the same backbone (ST-Tiny) but are trained jointly without any denoising diffusion process.

Table VII reports the performance comparison between PowerDiffuser and this nondiffusion hybrid baseline in ISMCBT-Tiny (Task310). PowerDiffuser consistently outperforms it across all evaluation metrics. This confirms that the integration of contrastive and reconstruction learning within the diffusion framework provides synergistic benefits beyond simply combining the two objectives.

In PowerDiffuser, the contrastive loss and reconstruction loss share a common diffusion-based representation pathway, whereas in the dual-branch MAE + SimCLR baseline, the two

TABLE VIII
COMPARISON OF COMPUTATIONAL EFFICIENCY BETWEEN POWERDIFFUSER AND BASELINES IN ETTh

Method	Training Time / Epoch (s)	Inference Time / Batch (ms)	Peak GPU Memory (GB)
CPC	24.7	9.6	1.7
TS-TCC	33.5	14.1	2.4
TimeMAE	46.1	30.0	5.1
CRT	60.7	76.0	6.6
PowerDiffuser	240.8	29.7	4.6

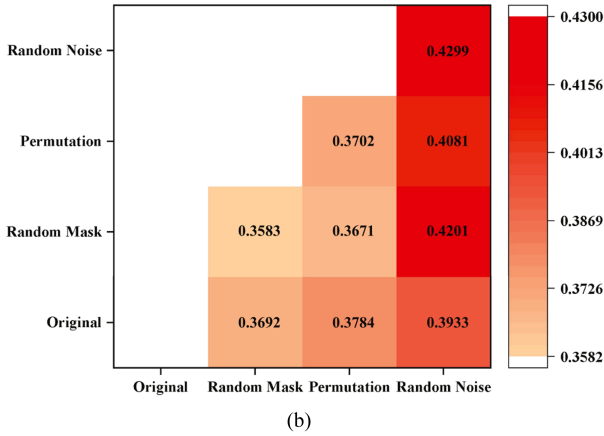
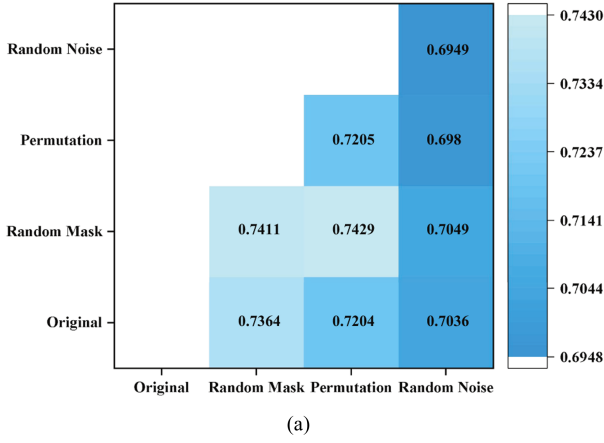


Fig. 9. Visualization of the impact of different feature enhancement combinations on performance. Combinations within the same method use different random seeds. It is worth noting that in both plots, lighter colors indicate better performance. (a) Comparison of pretraining loss values for different feature augmentation combinations on the ISMCBT dataset. (b) Comparison of different feature augmentation combinations on the downstream ISMCBT-tiny classification task (Task300).

objectives are executed in parallel and optimized independently. This may lead to conflicts in the representation space or mutual interference between objectives, making joint convergence more difficult. In contrast, the diffusion framework naturally provides a progressive learning mechanism, which we believe contributes to more stable and generalizable representation learning. Overall, we argue that PowerDiffuser’s joint objective design exhibits coupled consistency, rather than a structurally fragmented combination.

5) Computational Efficiency Analysis: We quantify the computational cost of PowerDiffuser and other self-supervised algorithms in Table VIII. Specifically, we perform pretraining on the ETTh dataset with a batch size of 64.

PowerDiffuser exhibits a relatively large training time, which is primarily attributed to the iterative nature of the diffusion process and the integration of both contrastive and reconstruction objectives at each step. In addition, our current implementation prioritizes modularity and clarity, which results in less runtime optimization compared to some mature baseline frameworks.

Despite the high training cost, the inference time of PowerDiffuser is 29.7 milliseconds per batch and the peak GPU memory usage is 4.6 GB. These values remain moderate and are even lower than those of single-objective self-supervised strategies. This efficiency is mainly due to the lightweight design of the ST feature extractor based on involution operations. Therefore, we consider the offline training time to be acceptable, and the inference efficiency indicates that PowerDiffuser has the potential for practical deployment.

VI. CONCLUSION

This article introduces PowerDiffuser, a self-supervised strategy specifically designed for power load signals. Built on the diffusion model framework, PowerDiffuser combines two mainstream self-supervised strategies, namely contrastive learning and reconstruction, effectively balancing global distribution and local information. We show that general time-series models often overfit in power load signal tasks and address this by developing two adaptable feature extractors, ST-Large and ST-Tiny, designed to process raw data of varying complexities. Furthermore, we offer insights into the generalization of the involution operator for 1-D data and the preprocessing of power load signals. Experimental results on real-world datasets confirm the effectiveness of our approach, while ablation studies emphasize the influence of different design choices on the overall architecture’s performance. We argue that a well-designed self-supervised strategy is essential for applying artificial intelligence technologies in the power load domain. Future work will focus on exploring more lightweight self-supervised strategies to mitigate the substantial computational costs associated with the diffusion process.

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